

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 27 August 2026, 8:06 AM**State** Finished**Completed on** Thursday, 17 September 2026, 9:40 AM**Time taken** 21 days 1 hour**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $arr[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```
1 |
2 | #include <stdio.h>
3 | #include <stdlib.h>
4 |
5 | int compare(const void *a, const void *b) {
6 |     long long x = *(const long long *)a;
7 |     long long y = *(const long long *)b;
8 |
9 |     if (x < y)
10 |         return -1;
11 |     if (x > y)
12 |         return 1;
13 |     return 0;
14 | }
15 |
16 | int main() {
17 |     int n;
18 |     scanf("%d", &n);
19 |
20 |     long long arr[n];
21 |
22 |     for (int i = 0; i < n; i++) {
23 |         scanf("%lld", &arr[i]);
24 |     }
25 |
26 |     // Greedy: sort in ascending order
27 |     qsort(arr, n, sizeof(long long), compare);
28 |
29 |     long long sum = 0;
30 |
31 |     for (int i = 0; i < n; i++) {
32 |         sum += arr[i] * i;
33 |     }
34 |
35 |     printf("%lld\n", sum);
36 |
37 |     return 0;
38 | }
39 |
```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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